# Document 4883

## Section 1

Mathematics is a field of study that discovers and organizes methods, theories and theorems that are developed and proved for the needs of empirical sciences and mathematics itself. There are many areas of mathematics, which include number theory (the study of numbers), algebra (the study of formulas and related structures), geometry (the study of shapes and spaces that contain them), analysis (the study of continuous changes), and set theory (presently used as a foundation for all mathematics).

Mathematics involves the description and manipulation of abstract objects that consist of either abstractions from nature or—in modern mathematics—purely abstract entities that are stipulated to have certain properties, called axioms. Mathematics uses pure reason to prove properties of objects, a proof consisting of a succession of applications of deductive rules to already established results. These results, called theorems, include previously proved theorems, axioms, and—in case of abstraction from nature—some basic properties that are considered true starting points of the theory under consideration.

Mathematics is essential in the natural sciences, engineering, medicine, finance, computer science, and the social sciences. Although mathematics is extensively used for modeling phenomena, the fundamental truths of mathematics are independent of any scientific experimentation. Some areas of mathematics, such as statistics and game theory, are developed in close correlation with their applications and are often grouped under applied mathematics. Other areas are developed independently from any application (and are therefore called pure mathematics) but often later find practical applications.

Historically, the concept of a proof and its associated mathematical rigour first appeared in Greek mathematics, most notably in Euclid's Elements. Since its beginning, mathematics was primarily divided into geometry and arithmetic (the manipulation of natural numbers and fractions), until the 16th and 17th centuries, when algebra and infinitesimal calculus were introduced as new fields. Since then, the interaction between mathematical innovations and scientific discoveries has led to a correlated increase in the development of both. At the end of the 19th century, the foundational crisis of mathematics led to the systematization of the axiomatic method, which heralded a dramatic increase in the number of mathematical areas and their fields of application. The contemporary Mathematics Subject Classification lists more than sixty first-level areas of mathematics.

Before the Renaissance, mathematics was divided into two main areas: arithmetic, regarding the manipulation of numbers, and geometry, regarding the study of shapes. Some types of pseudoscience, such as numerology and astrology, were not then clearly distinguished from mathematics.

During the Renaissance, two more areas appeared. Mathematical notation led to algebra which, roughly speaking, consists of the study and the manipulation of formulas. Calculus, consisting of the two subfields differential calculus and integral calculus, is the study of

continuous functions, which model the typically nonlinear relationships between varying quantities, as represented by variables. This division into four main areas—arithmetic, geometry, algebra, and calculus—endured until the end of the 19th century. Areas such as celestial mechanics and solid mechanics were then studied by mathematicians, but now are considered as belonging to physics. The subject of combinatorics has been studied for much of recorded history, yet did not become a separate branch of mathematics until the seventeenth century.
At the end of the 19th century, the foundational crisis in mathematics and the resulting systematization of the axiomatic method led to an explosion of new areas of mathematics. The 2020 Mathematics Subject Classification contains no less than sixty-three first-level areas. Some of these areas correspond to the older division, as is true regarding number theory (the modern name for higher arithmetic) and geometry. Several other first-level areas have "geometry" in their names or are otherwise commonly considered part of geometry. Algebra and calculus do not appear as first-level areas but are respectively split into several first-level areas. Other first-level areas emerged during the 20th century or had not previously been considered as mathematics, such as mathematical logic and foundations.

Number theory began with the manipulation of numbers, that is, natural numbers

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Number theory was once called arithmetic, but nowadays this term is mostly used for numerical calculations. Number theory dates back to ancient Babylon and probably China. Two prominent early number theorists were Euclid of ancient Greece and Diophantus of Alexandria. The modern study of number theory in its abstract form is largely attributed to Pierre de Fermat and Leonhard Euler. The field came to full fruition with the contributions of Adrien-Marie Legendre and Carl Friedrich Gauss.
Many easily stated number problems have solutions that require sophisticated methods, often from across mathematics. A prominent example is Fermat's Last Theorem. This conjecture was stated in 1637 by Pierre de Fermat, but it was proved only in 1994 by Andrew Wiles, who used tools including scheme theory from algebraic geometry, category theory, and homological algebra. Another example is Goldbach's conjecture, which asserts that every even integer greater than 2 is the sum of two prime numbers. Stated in 1742 by Christian Goldbach, it remains unproven despite considerable effort.
Number theory includes several subareas, including analytic number theory, algebraic number theory, geometry of numbers (method oriented), Diophantine analysis, and transcendence theory (problem oriented).

## Section 2

Geometry is one of the oldest branches of mathematics. It started with empirical recipes concerning shapes, such as lines, angles and circles, which were developed mainly for the needs of surveying and architecture, but has since blossomed out into many other subfields. A fundamental innovation was the ancient Greeks' introduction of the concept of proofs, which require that every assertion must be proved. For example, it is not sufficient to verify by measurement that, say, two lengths are equal; their equality must be proven via reasoning from previously accepted results (theorems) and a few basic statements. The basic statements are not subject to proof because they are self-evident (postulates), or are part of the definition of the subject of study (axioms). This principle, foundational for all

mathematics, was first elaborated for geometry, and was systematized by Euclid around 300 BC in his book Elements.
The resulting Euclidean geometry is the study of shapes and their arrangements constructed from lines, planes and circles in the Euclidean plane (plane geometry) and the three-dimensional Euclidean space.
Euclidean geometry was developed without change of methods or scope until the 17th century, when René Descartes introduced what is now called Cartesian coordinates. This constituted a major change of paradigm: Instead of defining real numbers as lengths of line segments (see number line), it allowed the representation of points using their coordinates, which are numbers. Algebra (and later, calculus) can thus be used to solve geometrical problems. Geometry was split into two new subfields: synthetic geometry, which uses purely geometrical methods, and analytic geometry, which uses coordinates systemically.
Analytic geometry allows the study of curves unrelated to circles and lines. Such curves can be defined as the graph of functions, the study of which led to differential geometry. They can also be defined as implicit equations, often polynomial equations (which spawned algebraic geometry). Analytic geometry also makes it possible to consider Euclidean spaces of higher than three dimensions.
In the 19th century, mathematicians discovered non-Euclidean geometries, which do not follow the parallel postulate. By questioning that postulate's truth, this discovery has been viewed as joining Russell's paradox in revealing the foundational crisis of mathematics. This aspect of the crisis was solved by systematizing the axiomatic method, and adopting that the truth of the chosen axioms is not a mathematical problem. In turn, the axiomatic method allows for the study of various geometries obtained either by changing the axioms or by considering properties that do not change under specific transformations of the space.
Today's subareas of geometry include:

Projective geometry, introduced in the 16th century by Girard Desargues, extends Euclidean geometry by adding points at infinity at which parallel lines intersect. This simplifies many aspects of classical geometry by unifying the treatments for intersecting and parallel lines.
Affine geometry, the study of properties relative to parallelism and independent from the concept of length.
Differential geometry, the study of curves, surfaces, and their generalizations, which are defined using differentiable functions.
Manifold theory, the study of shapes that are not necessarily embedded in a larger space.
Riemannian geometry, the study of distance properties in curved spaces.
Algebraic geometry, the study of curves, surfaces, and their generalizations, which are defined using polynomials.
Topology, the study of properties that are kept under continuous deformations.
Algebraic topology, the use in topology of algebraic methods, mainly homological algebra.
Discrete geometry, the study of finite configurations in geometry.
Convex geometry, the study of convex sets, which takes its importance from its applications in optimization.
Complex geometry, the geometry obtained by replacing real numbers with complex numbers.

Algebra is the art of manipulating equations and formulas. Diophantus (3rd century) and al-Khwarizmi (9th century) were the two main precursors of algebra. Diophantus solved some equations involving unknown natural numbers by deducing new relations until he obtained the solution. Al-Khwarizmi introduced systematic methods for transforming equations, such as moving a term from one side of an equation into the other side. The term algebra is derived from the Arabic word al-jabr meaning 'the reunion of broken parts' that he used for naming one of these methods in the title of his main treatise.

Algebra became an area in its own right only with François Viète (1540–1603), who introduced the use of variables for representing unknown or unspecified numbers. Variables allow mathematicians to describe the operations that have to be done on the numbers represented using mathematical formulas.

Until the 19th century, algebra consisted mainly of the study of linear equations (presently linear algebra), and polynomial equations in a single unknown, which were called algebraic equations (a term still in use, although it may be ambiguous). During the 19th century, mathematicians began to use variables to represent things other than numbers (such as matrices, modular integers, and geometric transformations), on which generalizations of arithmetic operations are often valid. The concept of algebraic structure addresses this, consisting of a set whose elements are unspecified, of operations acting on the elements of the set, and rules that these operations must follow. The scope of algebra thus grew to include the study of algebraic structures. This object of algebra was called modern algebra or abstract algebra, as established by the influence and works of Emmy Noether, and popularized by Van der Waerden's book Moderne Algebra.

Some types of algebraic structures have useful and often fundamental properties, in many areas of mathematics. Their study became autonomous parts of algebra, and include:

## Section 3

group theory

field theory

vector spaces, whose study is essentially the same as linear algebra

ring theory

commutative algebra, which is the study of commutative rings, includes the study of polynomials, and is a foundational part of algebraic geometry

homological algebra

Lie algebra and Lie group theory

Boolean algebra, which is widely used for the study of the logical structure of computers

The study of types of algebraic structures as mathematical objects is the purpose of universal algebra and category theory. The latter applies to every mathematical structure (not only algebraic ones). At its origin, it was introduced, together with homological algebra for allowing the algebraic study of non-algebraic objects such as topological spaces; this particular area of application is called algebraic topology.

Calculus, formerly called infinitesimal calculus, was introduced independently and simultaneously by 17th-century mathematicians Newton and Leibniz. It is fundamentally the study of the relationship between variables that depend continuously on each other. Calculus was expanded in the 18th century by Euler with the introduction of the concept of a function and many other results. Presently, "calculus" refers mainly to the elementary part of this theory, and "analysis" is commonly used for advanced parts.
Analysis is further subdivided into real analysis, where variables represent real numbers, and complex analysis, where variables represent complex numbers. Analysis includes many subareas shared by other areas of mathematics which include:

Multivariable calculus
Functional analysis, where variables represent varying functions
Integration, measure theory and potential theory, all strongly related with probability theory on a continuum
Ordinary differential equations
Partial differential equations
Numerical analysis, mainly devoted to the computation on computers of solutions of ordinary and partial differential equations that arise in many applications

## Section 4

Discrete mathematics, broadly speaking, is the study of individual, countable mathematical objects. An example is the set of all integers. Because the objects of study here are discrete, the methods of calculus and mathematical analysis do not directly apply.
Algorithms—especially their implementation and computational complexity—play a major role in discrete mathematics.
The four color theorem and optimal sphere packing were two major problems of discrete mathematics solved in the second half of the 20th century. The P versus NP problem, which remains open to this day, is also important for discrete mathematics, since its solution would potentially impact a large number of computationally difficult problems.
Discrete mathematics includes:

Combinatorics, the art of enumerating mathematical objects that satisfy some given constraints. Originally, these objects were elements or subsets of a given set; this has been extended to various objects, which establishes a strong link between combinatorics and other parts of discrete mathematics. For example, discrete geometry includes counting configurations of geometric shapes.
Graph theory and hypergraphs
Coding theory, including error correcting codes and a part of cryptography
Matroid theory
Discrete geometry
Discrete probability distributions
Game theory (although continuous games are also studied, most common games, such as chess and poker are discrete)

Discrete optimization, including combinatorial optimization, integer programming, constraint programming

The two subjects of mathematical logic and set theory have belonged to mathematics since the end of the 19th century. Before this period, sets were not considered to be mathematical objects, and logic, although used for mathematical proofs, belonged to philosophy and was not specifically studied by mathematicians.

Before Cantor's study of infinite sets, mathematicians were reluctant to consider actually infinite collections, and considered infinity to be the result of endless enumeration. Cantor's work offended many mathematicians not only by considering actually infinite sets but by showing that this implies different sizes of infinity, per Cantor's diagonal argument. This led to the controversy over Cantor's set theory. In the same period, various areas of mathematics concluded the former intuitive definitions of the basic mathematical objects were insufficient for ensuring mathematical rigour.

This became the foundational crisis of mathematics. It was eventually solved in mainstream mathematics by systematizing the axiomatic method inside a formalized set theory. Roughly speaking, each mathematical object is defined by the set of all similar objects and the properties that these objects must have. For example, in Peano arithmetic, the natural numbers are defined by "zero is a number", "each number has a unique successor", "each number but zero has a unique predecessor", and some rules of reasoning. This mathematical abstraction from reality is embodied in the modern philosophy of formalism, as founded by David Hilbert around 1910.

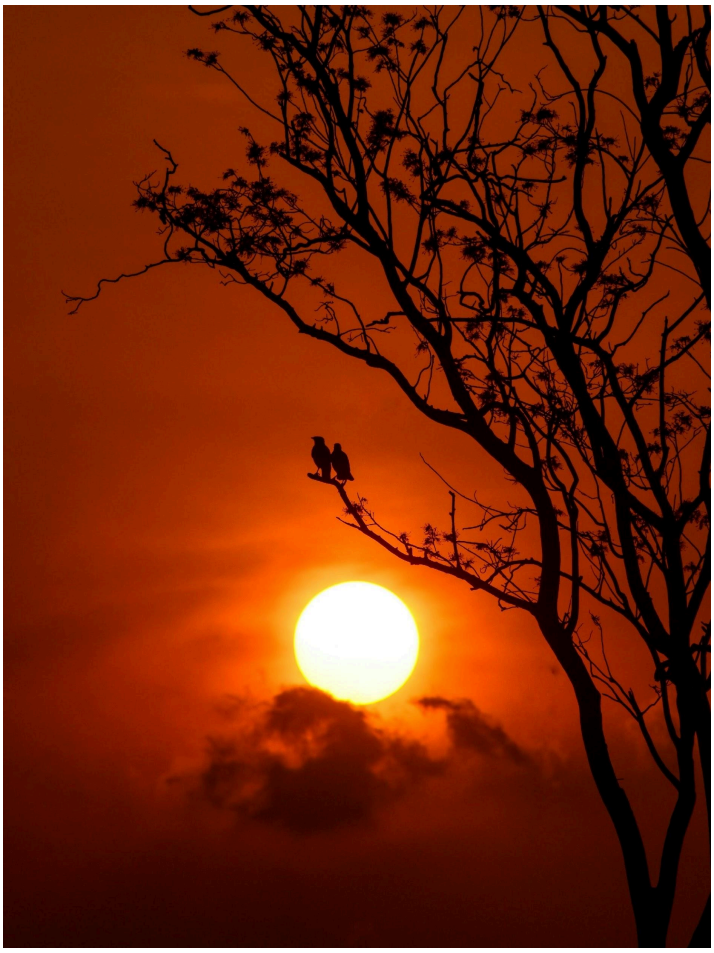
The "nature" of the objects defined this way is a philosophical problem that mathematicians leave to philosophers, even if many mathematicians have opinions on this nature, and use their opinion—sometimes called "intuition"—to guide their study and proofs. The approach allows considering "logics" (that is, sets of allowed deducing rules), theorems, proofs, etc. as mathematical objects, and to prove theorems about them. For example, Gödel's incompleteness theorems assert, roughly speaking that, in every consistent formal system that contains the natural numbers, there are theorems that are true (that is provable in a stronger system), but not provable inside the system. This approach to the foundations of mathematics was challenged during the first half of the 20th century by mathematicians led by Brouwer, who promoted intuitionistic logic, which explicitly lacks the law of excluded middle.

These problems and debates led to a wide expansion of mathematical logic, with subareas such as model theory (modeling some logical theories inside other theories), proof theory, type theory, computability theory and computational complexity theory. Although these aspects of mathematical logic were introduced before the rise of computers, their use in compiler design, formal verification, program analysis, proof assistants and other aspects of computer science, contributed in turn to the expansion of these logical theories.

## Section 5

The field of statistics is a mathematical application that is employed for the collection and processing of data samples, using procedures based on mathematical methods especially

probability theory. Statisticians generate data with random sampling or randomized experiments.
Statistical theory studies decision problems such as minimizing the risk (expected loss) of a statistical action, such as using a procedure in, for example, parameter estimation, hypothesis testing, and selecting the best. In these traditional areas of mathematical statistics, a statistical-decision problem is formulated by minimizing an objective function, like expected loss or cost, under specific constraints. For example, designing a survey often involves minimizing the cost of estimating a population mean with a given level of confidence. Because of its use of optimization, the mathematical theory of statistics overlaps with other decision sciences, such as operations research, control theory, and mathematical economics.

Computational mathematics is the study of mathematical problems that are typically too large for human, numerical capacity. Numerical analysis studies methods for problems in analysis using functional analysis and approximation theory; numerical analysis broadly includes the study of approximation and discretization with special focus on rounding errors. Numerical analysis and, more broadly, scientific computing also study non-analytic topics of mathematical science, especially algorithmic-matrix-and-graph theory. Other areas of computational mathematics include computer algebra and symbolic computation.

=== Etymology ===
The word mathematics comes from the Ancient Greek word máthēma (μάθημα), meaning 'something learned, knowledge, mathematics', and the derived expression mathēmatikḗ tékhnē (μαθηματικὴ τέχνη), meaning 'mathematical science'. It entered the English language during the Late Middle English period through French and Latin.
Similarly, one of the two main schools of thought in Pythagoreanism was known as the mathēmatikoi (μαθηματικοί)—which at the time meant "learners" rather than "mathematicians" in the modern sense. The Pythagoreans were likely the first to constrain the use of the word to just the study of arithmetic and geometry. By the time of Aristotle (384–322 BC) this meaning was fully established.
In Latin and English, until around 1700, the term mathematics more commonly meant "astrology" (or sometimes "astronomy") rather than "mathematics"; the meaning gradually changed to its present one from about 1500 to 1800. This change has resulted in several mistranslations: For example, Saint Augustine's warning that Christians should beware of mathematici, meaning "astrologers", is sometimes mistranslated as a condemnation of mathematicians.
The apparent plural form in English goes back to the Latin neuter plural mathematica (Cicero), based on the Greek plural ta mathēmatiká (τὰ μαθηματικά) and means roughly "all things mathematical", although it is plausible that English borrowed only the adjective mathematic(al) and formed the noun mathematics anew, after the pattern of physics and metaphysics, inherited from Greek. In English, the noun mathematics takes a singular verb. It is often shortened to maths or, in North America, math.